Janssen effect

Part A. Hydrostatics (0.5 points)

A.1 (0.2pt)

$p_1 =$

A.2 (0.3pt)

$p(z) =$

Part B. Janssen effect in a cylindrical vessel with vertical walls (2.9 points)

B.1 (0.5pt)

Equation:

B.2 (1.2pt)

$A =$

$B =$

$\lambda =$

B.3 (1.2pt)

$\rho =$

$K =$

Part C. Conical vessel (2.3 points)

C.1 (0.1pt)

$r(z) =$

C.2 (0.6pt)

Equation:

C.3 (0.8pt)

$A =$

$\gamma =$

$\beta =$

C.4 (0.8pt)

Graph:

$\frac{P_{\text{max}}}{\rho g z_{\text{max}}} =$

Part D. What if the vessel is lifted? (1.5 points)

D.1 (0.4pt)

Equation:

D.2 (0.3pt)

$A =$

$B =$

$\lambda =$

D.3 (0.8pt)

$H\ =$

Part E. Fluid outflow through the wall (1.8 points)

E.1 (0.2pt)

Equation:

E.2 (0.1pt)

$W_1 =$

E.3 (0.3pt)

$A\ =$

E.4 (0.3pt)

$p_2 =$

E.5 (0.3pt)

$v(h)\ =$

E.6 (0.6pt)

$\tau =$

Part F. "Flow" of granules through a wall (1 point)

F.1 (0.2pt)

$\theta_r =$

F.2 (0.8pt)

$q\ =$

$n\ =$